

## PROJECT PROGRESS REPORT

- Project Name: AirVisionGuard
- Reporting Period: July 2026      Date of Submission: July 30, 2026
- GitHub: <https://github.com/ThunGg/AirVisionGuard>
- Team Information:

| Name             | Matric number | NTU mail                 |
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### 1. Summary

- Overall Status: On Track
- Abstract: The desktop application is fully functional and operational, with deployment packaging currently pending. Current efforts are now focused on refining and optimizing the AI models.

### 2. Completed Deliverables

- Pipeline Implementation: Successfully deployed the end-to-end processing pipeline: Camera Input → Image Acquisition → Face Detection → Face Recognition (Embedding generation followed by Cosine Similarity evaluation against anchor references).
- System Infrastructure: Finalized the backend architecture and user interface (UI) development.
- Optimization: Conducted experiments focused on minimizing model power consumption.

### 3. Ongoing Tasks & Next Steps

- Model Efficiency Optimization: Researching and implementing optimization techniques to minimize model power consumption and achieve ultra-low energy overhead.
- We apply knowledge distillation to train a compressed student model (~75% parameter scale) against the previous model.
- During inference, Convolutional and Batch Normalization layers are merged into a single layer, reducing memory bandwidth usage and overall computational latency.

### 4. Acknowledgments

- We express our gratitude to SEP and iLab for their support throughout this project phase.